

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

WARNING

If natural gas leaks are present, do not store the equipment inside or in any area that is covered. Severe personal injury can result from asphyxiation or explosion.

CAUTION

Clean any external surfaces of fuel metering valve before removal. Dirt or contaminants can damage the fuel system.

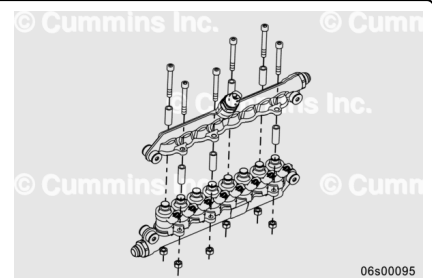
- Turn off the vehicle's main fuel supply valve.

- Operate the engine at low idle until the engine shuts down.
- Disconnect the batteries. See equipment manufacturer service information.
- Clean any external surfaces of the fuel metering valve.
- Disconnect the wiring harness from the engine gas temperature/pressure sensor and fuel metering valve injectors. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html)
- Remove the fuel metering valve. Refer to Procedure 005-242 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-242.html)

Remove

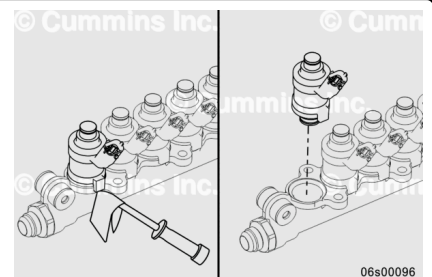
Remove the six capscrews and nuts.

Remove the six spacers between the nuts and capscrews.



Remove the injector gently.

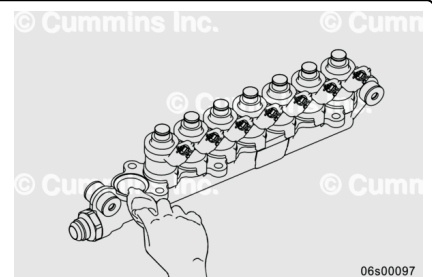
Wrap the screwdriver head (size 5 mm) with cloth, and use it leveraging the injector tank to remove the injector from the bore.



Clean and Inspect for Reuse

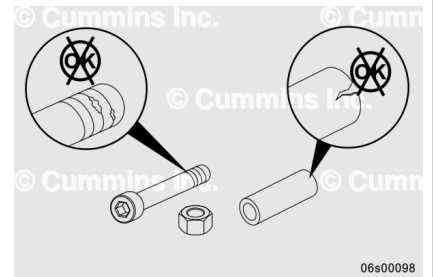
Clean the injector installation bores in the fuel metering valve block with clean cloth.

Inspect the bores for scratches, galling, burrs, and other damage. Replace the fuel metering valve block if damage is found.



Clean the nuts, spacers, and capscrews with mild solvent or n-heptane cleaner.

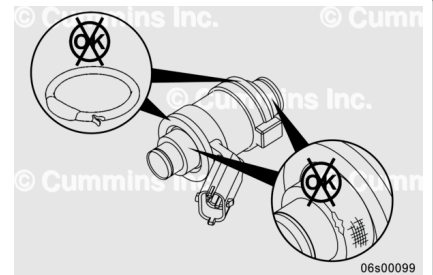
Inspect the nuts, spacers, and capscrews for nicks, cracks, burrs, and fretting. Replace if damage is found.



Inspect the injector contacting surface for nicks, scratches, galling, uneven wear, and other damage.

Inspect the o-rings for damage.

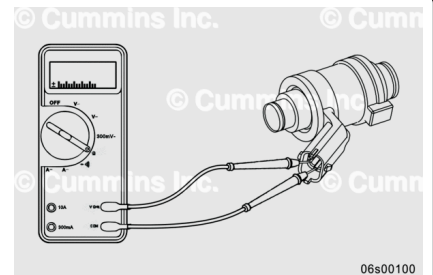
Replace the injector if damage is found.



Measure the fuel metering valve injector resistance.

The minimum resistance is 1.0 ohms.

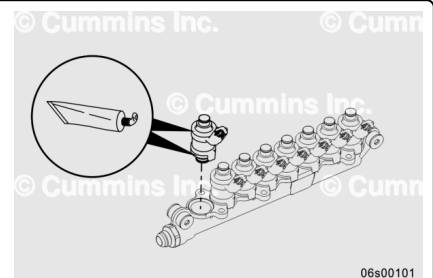
The maximum resistance is 3.5 ohms.



Install

Apply little clean oil to the injector o-rings and the sealing surfaces in the injector bore.

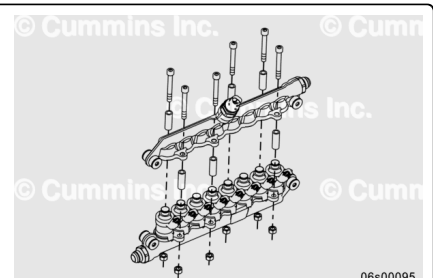
Insert the injectors into the bore. Ensure the fuel injector boss is locked into the groove of the housing. Make sure the electrical connector faces toward the outer side of the fuel metering valve.



Install the six capscrews, spacers, and nuts.

Tighten the nuts.

Torque Value: 17 n•m [150 in-lb]



Finishing Steps

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WARNING

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- Install the fuel metering valve. Refer to Procedure 005-242 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-242.html)
- Connect the engine wiring harness with the engine gas temperature/pressure sensor and fuel metering valve injectors. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html)
- Connect the batteries. See equipment manufacturer service information.
- Slowly open the vehicle's main fuel supply valve.
- Operate the engine and check for leaks.